

KGAGUELO MMAKOLA

Junior Full-Stack Data Scientist | Software Engineer | Business Analyst

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Portfolio: <https://15kay.github.io/mmk/>

Nationality: South African | Drivers Licence: Code 10

PROFESSIONAL SUMMARY

Versatile and results-driven Data Scientist, Software Engineer, and Business Analyst with strong experience in analytics, backend systems, AI automation, and database-driven solutions. Skilled in Python, Java, C#, PHP, JavaScript, and SQL, with the ability to build full-stack systems and generate insights from complex data. Recognized by Walter Sisulu University as **Top Tutor & Best Student (2023)** for outstanding performance, leadership, and innovation.

Delivers systems that solve real business problems, automate workflows, support academic decisions, and improve organizational efficiency.

TECHNICAL SKILLS

Programming & Scripting

- **Python** – Data analysis, automation, Pandas, NumPy, ML basics
- **Java** – OOP, backend logic, algorithms, teaching programming
- **C#** – .NET backend, CRUD systems, SQL integration
- **JavaScript** – Web interactivity, API calls, frontend logic
- **PHP** – Web server-side development, backend for web systems
- **Bash** – Automation scripts, Linux operations

Databases

- MySQL
- SQL Server
- PostgreSQL
- Database Design
- Queries

- Normalization

Tools & Technologies

- Git & GitHub
- Jupiter Notebook
- n8n AI Automation
- VS Code
- Linux
- REST APIs

Business Analysis & Project Skills

- Requirements Gathering
- ERDs
- UML
- Use Cases
- Process Mapping
- Report Writing
- Data Interpretation
- Stakeholder Communication

HOW I USE EACH LANGUAGE

Python

- Data cleaning, transformation, analytics, and automation
- Creating risk prediction insights for WSU advisors
- Running ML basics for academic and health datasets
- Generating reports and dashboards

Java

- Backend logic for student and admin systems
- OOP teaching (classes, inheritance, arrays, exceptions)
- Building console apps and solving algorithmic problems

C#

- Full backend system development using .NET

- CRUD apps, admin dashboards, authentication
- Connecting SQL Server databases to enterprise systems

JavaScript

- Creating dynamic web pages and validation logic
- Handling API requests between frontend and backend
- Supporting n8n automation flows

PHP

- Building **server-side logic for web applications**
- Managing form submissions, sessions, and backend workflows
- Integrating MySQL databases with web systems
- Creating responsive portals and simple dashboards

SQL

- Writing queries for analytics, reports, and backends
- Designing databases for applications
- Managing student data, bookings, products, and system records

Bash

- Running scripts, setting up environments
- Lightweight automation and Linux operations

WORK EXPERIENCE

Walter Sisulu University — Junior Data Analytics

2023 – 2025

- Analyzed student academic performance data to identify at-risk learners.
- Supported the **WSU Pre-Admission Risk Prediction Model**.
- Generated automated weekly & monthly analytics reports for academic advisors.
- Assisted with system requirements, documentation, and process improvement.
- Contributed to data reporting for faculty initiatives.

Walter Sisulu University — Development Software Tutor

2021, 2022, 2023

- Taught programming languages (Java, SQL, Python, C#, JavaScript, PHP basics).
- Assisted students with algorithms, flowcharts, pseudocode, and system design.
- Delivered high-quality tutoring sessions and project support.
- Won **Top Tutor Award & Best Student (2023)**.

MAJOR PROJECTS

Data Science

- **WSU Student Risk Analytics System** – Cleaned, analyzed, and visualized academic risk data.
- **DIRISA 2024 Challenge** – Contributed to disease early detection modelling.
- **Heart Disease Classification (ML)**

Software Engineering

- **Car Parts & Booking Management System (C# + SQL)** — Admin dashboard + appointments.
- **Curriculum Sync** — AI-driven dynamic curriculum generator.
- **Student Document Upload Portal (PHP + MySQL)** Secure upload & retrieval.

Business Analysis

- **Campus Network Upgrade Project** — Objectives, dependencies, stakeholder documentation.
- **Illegal Electricity Detection Prototype** — Requirements analysis & system modelling.

EDUCATION

Diploma in Information Technology (Application Development)

Walter Sisulu University | *2021 – 2024*

Advanced IT Application

Walter Sisulu University | *2025 – 2025*

Matric (Grade 12)

Luthuli High School | *2020*

LEADERSHIP & COMMUNITY INVOLVEMENT

- Secretary — **ZCC Student Fellowship (SF) Buffalo City Campus**
- Chairperson — **ZCC Student Fellowship (SF) Buffalo City Campus**
- Class representative – Walter Sisulu University(4years)
- Mentor — IT, analytics, and programming support for peers

REFERENCES

Mr. Sizani

Tutor Coordinator – Walter Sisulu University
061 710 0586

Mr. Mapatagane

Director – Walter Sisulu University
065 542 9854